

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                         |
|----------------------|-------------------------|
| United States Patent | 12418005                |
| Kind Code            | B2                      |
| Date of Patent       | September 16, 2025      |
| Inventor(s)          | Liu; Tzuan-Horng et al. |

---

### Semiconductor package and manufacturing method thereof

---

#### Abstract

A semiconductor package and a manufacturing method are provided. The semiconductor package includes a first die disposed on and electrically coupled to a first redistribution structure and laterally covered by a first insulating encapsulation, a second die disposed over the first die and laterally covered by a second insulating encapsulation, a second redistribution structure interposed between and electrically coupled to the first and second dies, a third redistribution structure disposed on the second die and opposite to the second redistribution structure, and at least one thermal-dissipating feature embedded in a dielectric layer of the third redistribution structure and electrically isolated from a patterned conductive layer of the third redistribution structure through the dielectric layer. Through substrate vias of the first die are physically connected to the second redistribution structure or the first redistribution structure. The thermal-dissipating feature is thermally coupled to a back surface of the second die.

---

**Inventors:** Liu; Tzuan-Horng (Taoyuan, TW), Tsai; Hao-Yi (Hsinchu, TW), Chuang; Kris Lipu (Hsinchu, TW), Pan; Hsin-Yu (Taipei, TW)

**Applicant:** Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

**Family ID:** 1000008821474

**Assignee:** Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

**Appl. No.:** 17/880687

**Filed:** August 04, 2022

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20240047436 A1          | Feb. 08, 2024           |

---

#### Publication Classification

**Int. Cl.:** H01L25/10 (20060101); H01L25/00 (20060101); H01L25/18 (20230101); H01L23/00 (20060101)

**U.S. Cl.:**

**CPC** H01L25/105 (20130101); H01L25/18 (20130101); H01L25/50 (20130101); H01L24/16 (20130101); H01L24/20 (20130101); H01L24/32 (20130101); H01L24/73 (20130101); H01L2224/16227 (20130101); H01L2224/214 (20130101); H01L2224/32225 (20130101); H01L2224/32245 (20130101); H01L2224/73204 (20130101); H01L2224/73259 (20130101); H01L2224/73267 (20130101); H01L2225/1035 (20130101); H01L2225/1041 (20130101); H01L2225/1058 (20130101); H01L2225/1094 (20130101)

## Field of Classification Search

**CPC:** H01L (25/105); H01L (25/18); H01L (25/50); H01L (24/16); H01L (24/20); H01L (24/32); H01L (24/73); H01L (2224/16227); H01L (2224/214); H01L (2224/32225); H01L (2224/32245); H01L (2224/73204); H01L (2224/73259); H01L (2224/73267); H01L (2225/1035); H01L (2225/1041); H01L (2225/1058); H01L (2225/1094)

---

## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name | U.S. Cl.    | CPC             |
|--------------|-------------|---------------|-------------|-----------------|
| 9000584      | 12/2014     | Lin et al.    | N/A         | N/A             |
| 9048222      | 12/2014     | Hung et al.   | N/A         | N/A             |
| 9048233      | 12/2014     | Wu et al.     | N/A         | N/A             |
| 9064879      | 12/2014     | Hung et al.   | N/A         | N/A             |
| 9111949      | 12/2014     | Yu et al.     | N/A         | N/A             |
| 9263511      | 12/2015     | Yu et al.     | N/A         | N/A             |
| 9281254      | 12/2015     | Yu et al.     | N/A         | N/A             |
| 9368460      | 12/2015     | Yu et al.     | N/A         | N/A             |
| 9372206      | 12/2015     | Wu et al.     | N/A         | N/A             |
| 9496189      | 12/2015     | Yu et al.     | N/A         | N/A             |
| 2010/0133704 | 12/2009     | Marimuthu     | 257/E23.116 | H01L<br>23/3128 |
| 2017/0098633 | 12/2016     | Kumar         | N/A         | H01L<br>23/3114 |
| 2019/0148262 | 12/2018     | Pei           | 257/713     | H01L 25/105     |
| 2020/0006196 | 12/2019     | Lin           | N/A         | H01L<br>23/5389 |
| 2020/0194331 | 12/2019     | Kim           | N/A         | H01L 24/08      |
| 2021/0082783 | 12/2020     | Kim           | N/A         | H01L<br>23/3677 |

### FOREIGN PATENT DOCUMENTS

| Patent No. | Application Date | Country | CPC |
|------------|------------------|---------|-----|
| 201916280  | 12/2018          | TW      | N/A |
| 202029425  | 12/2019          | TW      | N/A |
| 202109794  | 12/2020          | TW      | N/A |

## OTHER PUBLICATIONS

“Office Action of Taiwan Counterpart Application”, issued on Nov. 13, 2023, p. 1-p. 10. cited by applicant

---

*Primary Examiner:* Toledo; Fernando L

*Assistant Examiner:* Prasad; Neil R

*Attorney, Agent or Firm:* JCIPRNET

---

## Background/Summary

### BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced a fast-paced growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component or line that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. However, such scaling has also introduced high heat density and poor thermal dissipation performance to the semiconductor package. Increased heat density in three-dimensional system can lead to electromigration and reliability issues.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIGS. 1A-1K are schematic cross-sectional views illustrating various stages of a manufacturing method of a semiconductor package, in accordance with some embodiments.
- (3) FIG. 2A is a schematic top view illustrating a configuration of thermal-dissipating vias, in accordance with some embodiments.
- (4) FIG. 2B is a schematic top view illustrating a configuration of a thermal-dissipating pattern, in accordance with some embodiments.
- (5) FIG. 2C is a schematic top view illustrating a combined configuration of the thermal-dissipating pattern and the thermal-dissipating vias, in accordance with some embodiments.
- (6) FIG. 2D is a schematic top view illustrating a configuration of conductive terminals and thermal-dissipating bumps, in accordance with some embodiments.
- (7) FIGS. 3-4, and 6A are schematic cross-sectional views illustrating various semiconductor packages, in accordance with some embodiments.
- (8) FIG. 5A is a schematic top view illustrating a configuration of a thermal-dissipating pattern, in accordance with some embodiments.
- (9) FIG. 5B is a schematic top view illustrating a combined configuration of multi-leveled thermal-

dissipating patterns, in accordance with some embodiments.

(10) FIG. 6B is a schematic top view illustrating a configuration of the thermal-dissipating feature, the first die, and the second die of the semiconductor package of FIG. 6A, in accordance with some embodiments.

(11) FIGS. 7A-7I are schematic cross-sectional views illustrating various stages of a manufacturing method of a semiconductor package, in accordance with some embodiments.

#### DETAILED DESCRIPTION

(12) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(13) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. In addition, terms, such as “first,” “second,” “third,” “fourth” and the like, may be used herein for ease of description to describe similar or different element(s) or feature(s) as illustrated in the figures, and may be used interchangeably depending on the order of the presence or the contexts of the description.

(14) Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

(15) FIGS. 1A-1K are schematic cross-sectional views illustrating various stages of a manufacturing method of a semiconductor package, FIG. 2A is a schematic top view illustrating a configuration of thermal-dissipating vias, FIG. 2B is a schematic top view illustrating a configuration of a thermal-dissipating pattern, FIG. 2C is a schematic top view illustrating a combined configuration of the thermal-dissipating pattern and the thermal-dissipating vias, and FIG. 2D is a schematic top view illustrating a configuration of conductive terminals and thermal-dissipating bumps, in accordance with some embodiments.

(16) Referring to FIG. 1A, a first redistribution structure **110** is formed over a first temporary carrier **51**. The first temporary carrier **51** may be made of a material such as silicon, polymer, polymer composite, metal foil, ceramic, glass, glass epoxy, tape, or other suitable material for structural support. In some embodiments, an adhesive layer (not shown) is formed on the first temporary carrier **51** before the first redistribution structure **110** is formed. The adhesive layer may be detached from the first temporary carrier **51** by, e.g., shining an ultra-violet (UV) light on the first temporary carrier **51** in a subsequent carrier de-bonding process. For example, the adhesive

layer is a light-to-heat-conversion (LTHC) coating layer or the like.

(17) The first redistribution structure **110** may include one or more first patterned conductive layer(s) **112** formed in one or more first dielectric layer(s) **114**. In some embodiments, the first dielectric layer **114** is formed of a polymer, such as polybenzoxazole (PBO), polyimide, benzocyclobutene (BCB), or the like; a nitride such as silicon nitride; an oxide such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), or the like. In some embodiments, the first patterned conductive layer **112** includes conductive lines, conductive vias, and conductive pads and may be formed of a suitable conductive material such as copper, titanium, tungsten, aluminum, alloys, or the like.

(18) In some embodiments, the bottommost sublayer of the first dielectric layer **114** is formed and patterned over the first temporary carrier **51** by using lithography and etching or other suitable processes, and then the bottommost sublayer of the first patterned conductive layer **112** is formed on the top surface of the bottommost sublayer of the first dielectric layer **114** and in openings of the bottommost sublayer of the first dielectric layer **114**. The steps of forming a sublayer of the first dielectric layer **114** and forming a sublayer of the first patterned conductive layer **112** may be repeated to form the first redistribution structure **110**. It is noted that the number of sublayers of the first dielectric layer **114** and the first patterned conductive layer **112** in the first redistribution structure **110** may construe no limitation in the disclosure. Other methods of forming the first redistribution structure **110** are possible and fully intended to be included within the scope of the disclosure.

(19) In some embodiments, the first redistribution structure **110** includes a first surface **110a** and a second surface **110b** opposite to the first surface **110a** and facing the first temporary carrier **51**. For example, surfaces **1104b** and **1102b** of the bottommost sublayers of the first dielectric layer **114** and the first patterned conductive layer **112** on the second surface **110b** are substantially leveled (e.g., coplanar) with each other. In some embodiments, the first surface **110a** includes a top surface of the topmost sublayer of the first dielectric layer **114** and a top surface of the topmost sublayer of the first patterned conductive layer **112**, where the top surface of the topmost sublayer of the first patterned conductive layer **112** is protruded from the top surface of the topmost sublayer of the first dielectric layer **114**. For example, the topmost sublayer of the first patterned conductive layer **112** includes first pads **1121** and second pads **1122** surrounding the first pads **1121**. The pitch of the adjacent first pads **1121** may be less than that of the adjacent second pads **1122**. The density per unit area of the first pads **1121** may be denser than that of the second pads **1122**. The first and second pads **1121** and **1122** formed on the top surface of the topmost sublayer of the first dielectric layer **114** may be or include under bump metallization (UBM) structure for further electrical connection.

(20) Referring to FIG. 1B, first conductive pillars **120** are formed on the second pads **1122** of the first patterned conductive layer **112** of the first redistribution structure **110**. The first conductive pillars **120** may be formed by: forming a seed layer; forming a patterned photoresist over the seed layer, where each of the openings in the patterned photoresist corresponds to a location of the respective first conductive pillar **120** to be formed; filling the openings with an electrically conductive material such as copper using, e.g., plating or the like; removing the patterned photoresist using, e.g., an ashing or a stripping process; and removing portions of the seed layer on which the first conductive pillars **120** are not formed. Other methods for forming the first conductive pillars **120** are possible and fully intended to be included within the scope of the disclosure. By way of example and not limitation, a pitch **120P** between two adjacent first conductive pillars **120** may be about 60  $\mu\text{m}$ . Although other value(s) are also possible.

(21) In some embodiments, a first die **130'** is disposed over the first surface **110a** of the first redistribution structure **110** and electrically coupled to the first patterned conductive layer **112**. The first die **130'** may be surrounded by the first conductive pillars **120**. The first die **130'** may be cut from a semiconductor wafer (not shown). The first die **130'** may include a first semiconductor

substrate **131'** having a front surface **131a** and a back surface **131b'** opposite to the front surface **131a**, conductive vias **132'** embedded in the first semiconductor substrate **131'** and extending from the front surface **131a** toward the back surface **131b**. The first semiconductor substrate **131'** may include an elementary semiconductor (e.g., silicon or germanium in a crystalline, a polycrystalline, or an amorphous structure, etc.), a compound semiconductor (e.g., SiC, GaAs, GaP, InP, InAs, and/or InSb, etc.), an alloy semiconductor (e.g., SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, etc.), combinations thereof, or other suitable materials. In some embodiments, the first semiconductor substrate **131'** may be a compound semiconductor substrate having a multilayer structure or any suitable substrate. The conductive vias **132'** may include one or more conductive materials (e.g., cobalt, titanium, tungsten, copper, aluminum, tantalum, titanium nitride, tantalum nitride, gold, silver, another metal, a metal alloy, or combinations thereof).

(22) The first die **130'** may include a first device layer **133** having semiconductor devices (not shown) formed in/on the front surface **131a** of the first semiconductor substrate **131'** and electrically coupled to the conductive vias **132'**, a first interconnect structure **134** underlying the first device layer **133** and electrically coupled to the semiconductor devices of the first device layer **133**, and first die connectors **135** underlying and electrically coupled to the first interconnect structure **134**. The semiconductor devices in the first device layer **133** may be or include active devices (e.g., transistors, diodes, etc.) and/or passive devices (e.g., capacitors, resistors, inductors, etc.), or other suitable electrical devices. The first interconnect structure **134** may include one or more interconnect wiring layer(s) embedded in one or more interconnect dielectric layer(s), where the interconnect wiring layers are electrically coupled to the semiconductor devices in the first device layer **133** and the first die connectors **135**. The first die connectors **135** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the first die connectors **135** include metal pillars (e.g., a copper pillar) formed by a sputtering, printing, plating, CVD, or the like, with or without a solder cap thereon. The metal pillars may be solder-free and have substantially vertical sidewalls or tapered sidewalls.

(23) With continued reference to FIG. **1B**, the first die connectors **135** of the first die **130'** may be coupled to the first pads **1121** of the first redistribution structure **110** with a one-to-one correspondence. In some embodiments, first conductive joints **136**, e.g., solder joints, are formed to connect the metal pillars of the first die connectors **135** and the first pads **1121**. In some embodiments, a first underfill layer **UF1** is formed in a gap between the first die **130'** and the first redistribution structure **110** to surround the first conductive joints **136**, the first die connectors **135**, and the first pads **1121**. In some embodiments, the first underfill layer **UF1** extends upwardly to cover sidewalls of the first die **130'**. The first underfill layer **UF1** may be any acceptable material, such as a polymer, epoxy, molding underfill, or the like, and may be formed by dispensing or any other suitable method. In some embodiments, the first conductive pillars **120** are formed prior to the placement of the first die **130'**. Alternatively, the first die **130'** is coupled to the first redistribution structure **110** prior to the formation of the first conductive pillars **120**.

(24) Referring to FIG. **1C** and with reference to FIG. **1B**, a first insulating encapsulation **140** may be formed on the first redistribution structure **110** to cover the first conductive pillars **120**, the first underfill layer **UF1**, and the first die **130**. In some embodiments, the first insulating encapsulation **140** is a molding compound formed by a molding process. For example, the first insulating encapsulation **140** may include polymers (e.g., epoxy resins, phenolic resins, silicon-containing resins, or other suitable resins), dielectric materials, or other suitable materials. In some embodiments, the first insulating encapsulation **140** is made of a molding underfill material. In some embodiments, the first insulating encapsulation **140** includes inorganic fillers which can be added to optimize coefficient of thermal expansion (CTE) of the first insulating encapsulation **140**. The disclosure is not limited thereto.

(25) With continued reference to FIGS. **1C** and **1B**, in some embodiments, a layer of insulating

encapsulation material is formed to encapsulate the first conductive pillars **120**, the first underfill layer **UF1**, and the first die **130'**, and then a planarization process (e.g., chemical mechanical polishing (CMP), mechanical grinding, etching, a combination thereof, etc.) is performed on the insulating encapsulation material at least until the first conductive pillars **120** are accessibly exposed. In some embodiments, during the planarization process, the back surface **131b'** of the first semiconductor substrate **131'** of the first die **130'** is thinned down. In some embodiments, a portion of the back side of the first semiconductor substrate **131'** may be further removed such that the conductive vias **132'** are protruded from the back surface **131b** of the first semiconductor substrate **131** through etching or any suitable removal process. Subsequently, an isolation layer **137** may be formed on the back surface **131b** of the first semiconductor substrate **131** to laterally cover the conductive vias **132**. The isolation layer **137** may be made of low temperature polyimide or other suitable insulating material(s).

(26) In some embodiments, the first conductive pillars **120** that penetrate through the first insulating encapsulation **140** may be referred to as first through insulation vias (TIVs) **120**, and the conductive vias **132** that penetrate through the first semiconductor substrate **131** may be referred to as through substrate vias (TSVs) **132**. In some embodiments, the top surface **130t** of the first insulating encapsulation **140** is substantially leveled (or coplanar) with the top surfaces **120t** of the first TIVs **120**, within process variations. The top surface **140t** of the first insulating encapsulation **140** may also be substantially leveled (or coplanar) with the back surface **130t** of the first die **130**, within process variations, where the back surface **130t** of the first die **130** includes top surfaces of the TSVs **132** and the isolation layer **137**.

(27) Referring to FIG. 1D and with reference to FIG. 1C, a second redistribution structure **150** may be formed on the first TIVs **120**, the first die **130**, and the first insulating encapsulation **140**. The second redistribution structure **150** may include one or more second patterned conductive layer(s) **152** formed in one or more second dielectric layer(s) **154**. In some embodiments, the second dielectric layer **154** may include a polymer, such as PBO, polyimide, BCB, or the like; a nitride such as silicon nitride; an oxide such as silicon oxide, PSG, BSG, BPSG, or the like. The second dielectric layer **154** may be formed by a suitable deposition process, such as spin-coating, CVD, the like, or a combination thereof. In some embodiments, the second patterned conductive layer **152** includes conductive lines, conductive vias, and conductive pads and may be formed of a suitable conductive material such as copper, titanium, tungsten, aluminum, or the like.

(28) In some embodiments, the bottommost sublayer of the second dielectric layer **154** overlies the top surfaces of the first TIVs **120**, the first die **130**, and the first insulating encapsulation **140**, and the bottommost sublayer of the second patterned conductive layer **152** is in physical and electrical contact with the top surfaces of the first TIVs **120** and the TSVs **132** of the first die **130**. The second redistribution structure **150** may be electrically coupled to the first redistribution structure **110** through the first die **130** and/or the first TIVs **120**. In some embodiments, the topmost sublayer of the second patterned conductive layer **152** has first pads **1521** and second pads **1522** formed on the topmost sublayer of the second dielectric layer **154**, where the first pads **1521** are located right above the first die **130**, and the second pads **1522** surround the first pads **1521**. The pitch of the adjacent first pads **1521** may be less than that of the adjacent second pads **1522**. The density per unit area of the first pads **1521** may be denser than that of the second pads **1522**. The first and second pads **1521** and **1522** may be UBM structure for further electrical connection.

(29) Referring to FIG. 1E, second conductive pillars **160** are formed on the second pads **1522** of the second patterned conductive layer **152** of the second redistribution structure **150**. The second conductive pillars **160** may be formed by a method the same as (or similar to) the formation of the first conductive pillars **120** described in FIG. 1B. In some embodiments, a pitch **160P** between two adjacent second conductive pillars **160** is greater than the pitch **120P** between two adjacent first conductive pillars **120** (labeled in FIG. 1B). By way of example and not limitation, the pitch **160P** may be about 200  $\mu\text{m}$ . Although other value(s) are fully intended to be included within the scope of

the disclosure.

(30) In some embodiments, a second die **170** is disposed over the second redistribution structure **150** and electrically coupled to the second patterned conductive layer **152**. The second die **170** may be surrounded by the second conductive pillars **160**. The second die **170** may be cut from a semiconductor wafer (not shown). The second die **170** may include a second semiconductor substrate **171** having a front surface **171a** and a back surface **171b** opposite to the front surface **171a**. The material of the second semiconductor substrate **171** may be the same as or similar to that of the first semiconductor substrate **131** of the first die **130**. The second die **170** may include a second device layer **173** having semiconductor devices (not shown) and formed in/on the front surface **171a** of the second semiconductor substrate **171**, a second interconnect structure **174** underlying the second device layer **173** and electrically coupled to the semiconductor devices of the second device layer **173**, and second die connectors **175** underlying and electrically coupled to the second interconnect structure **174**.

(31) The semiconductor devices in the second device layer **173** may be or include active devices (e.g., transistors, diodes, etc.) and/or passive devices (e.g., capacitors, resistors, inductors, etc.), or other suitable electrical devices. The second interconnect structure **174** may include one or more interconnect wiring layer(s) embedded in one or more interconnect dielectric layer(s), where the interconnect wiring layers are electrically coupled to the semiconductor devices in the second device layer **173** and the second die connectors **175**. The second die connectors **175** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the second die connectors **175** include metal pillars (e.g., a copper pillar) formed by a sputtering, printing, plating, CVD, or the like, with or without a solder cap thereon. The metal pillars may be solder-free and have substantially vertical sidewalls or tapered sidewalls.

(32) With continued reference to FIG. **1E**, the second die connectors **175** of the second die **170** may be coupled to the first pads **1521** of the second redistribution structure **150** with a one-to-one correspondence. In some embodiments, second conductive joints **176**, e.g., solder joints, are formed to connect the metal pillars of the second die connectors **175** and the first pads **1521**. In some embodiments, a second underfill layer **UF2** is formed in a gap between the second die **170** and the second redistribution structure **150** to surround the second conductive joints **176**, the second die connectors **175**, and the first pads **1521**. In some embodiments, the second underfill layer **UF2** extends upwardly to cover sidewalls of the second die **170**. The second underfill layer **UF2** may be the same as or similar to the first underfill layer **UF1** described in FIG. **1B**. In some embodiments, the second conductive pillars **160** are formed prior to the placement of the second die **160**. Alternatively, the second die **160** is coupled to the second redistribution structure **150** prior to the formation of the second conductive pillars **160**.

(33) Referring to FIG. **1F**, a second insulating encapsulation **180** may be formed on the second redistribution structure **150** to cover the second conductive pillars **160**, the second underfill layer **UF2**, and the second die **170**. The second insulating encapsulation **180** may be the same as or similar to the first insulating encapsulation **140** described in FIG. **1C**. In some embodiments, a layer of insulating encapsulation material is formed to encapsulate the second conductive pillars **160**, the second underfill layer **UF2**, and the second die **170**, and then a planarization process (e.g., CMP, mechanical grinding, etching, a combination thereof, etc.) is performed on the insulating encapsulation material at least until the second conductive pillars **160** are accessibly exposed. During the planarization process, a portion of the back side of the second semiconductor substrate **171** of the second die **170** may be removed, such that the back surface **170b** of the second die **170** may be accessibly revealed by the second insulating encapsulation **180**. In some embodiments, the second conductive pillars **160** that penetrate through the second insulating encapsulation **160** may be referred to as second TIVs **160**. In some embodiments, the top surface **180t** of the second insulating encapsulation **180** is substantially leveled (or coplanar) with the top surfaces **160t** of the



second TIVs **160** and the back surface **170b** of the second die **170**, within process variations. (34) In some embodiments, the first die **130** and the second die **170** are of different types of dies. The thickness **170H** of the second die **170** may be greater than the thickness **130** of the first die **130**. The first die **130** and/or the second die **170** may be or include a logic die (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a system-on-a-chip (SoC), an application processor (AP), and a microcontroller); a power management die; a wireless and radio frequency (RF) die; a baseband (BB) die; a sensor die; a micro-electro-mechanical-system (MEMS) die; a signal processing die; a front-end die (e.g., an analog front-end (AFE) die); an application-specific integrated circuit (ASIC) die; a combination thereof; or the like. In alternative embodiments, the first die **130** and/or the second die **170** may be or include a memory die (e.g., a dynamic random-access memory (DRAM) die, a static random-access memory (SRAM) die, a resistive random-access memory (RRAM), a magneto-resistive random-access memory (MRAM), a NAND flash memory, a hybrid memory cube (HMC) module, a high bandwidth memory (HBM) module); a combination thereof; or the like. In alternative embodiments, the first die **130** and/or the second die **170** may be or include an artificial intelligence (AI) engine; a computing system (e.g., an AI server, a high-performance computing (HPC) system, a high-power computing device, a cloud computing system, a networking system, an edge computing system, a SoIC system, etc.); a combination thereof; or the like.

(35) Referring to FIG. **1G**, a third redistribution structure **190** may be formed on the second TIVs **160**, the second die **170**, and the second insulating encapsulation **180**. The third redistribution structure **190** may include one or more third patterned conductive layer(s) **192** formed in one or more third dielectric layer(s) **194**. In some embodiments, the third dielectric layer **194** includes a polymer, such as PBO, polyimide, BCB, or the like; a nitride such as silicon nitride; an oxide such as silicon oxide, PSG, BSG, BPSG, or the like. The third dielectric layer **194** may be formed by a suitable deposition process, such as spin-coating, CVD, the like, or a combination thereof. In some embodiments, the third patterned conductive layer **192** includes conductive lines, conductive vias, and conductive pads, and may be formed of a suitable conductive material such as copper, titanium, tungsten, aluminum, or the like.

(36) In some embodiments, the third patterned conductive layer **192** are formed within a circuit region **1901** of the third redistribution structure **190** directly above the second TIVs **160**. A region directly above the second die **170** may be a thermal-dissipating region **1902** which is free of the third patterned conductive layer **192**. For example, the thermal-dissipating region **1902** is surrounded by the circuit region **1901** and electrically isolated from the circuit region **1901** through the third dielectric layer **194**. The term “isolated” as used herein refers to a structure or layer that is not integrated with (e.g., part of) another structure, such as the patterned conductive layer, and it is disposed in a non-functional area in the third redistribution structure. In some embodiments, thermal-dissipating features **196** are formed within the thermal-dissipating region **1902** and may be electrically floating in the third redistribution structure **190**. The thermal-dissipating features **196** may include first thermal-dissipating vias **1961** directly landing on the back surface **170b** of the second die **170**, a first thermal-dissipating pattern **1962** overlying the first thermal-dissipating vias **1961**, second thermal-dissipating vias **1963** landing on the first thermal-dissipating pattern **1962**, and a second thermal-dissipating pattern **1964** overlying the second thermal-dissipating vias **1963**.

(37) With continued reference to FIG. **1G**, the first thermal-dissipating vias **1961**, the first thermal-dissipating pattern **1962**, the second thermal-dissipating vias **1963**, and the second thermal-dissipating pattern **1964** may include conductive material(s) such as copper, titanium, tungsten, aluminum, or the like. The first and second thermal-dissipating vias **1961** and **1963** may be formed during the same step of forming the conductive vias of the third patterned conductive layer **192**, and the first and second thermal-dissipating patterns **1962** and **1964** may be formed during the same step of forming the conductive pads and conductive lines of the third patterned conductive layer **192**. Although four levels of the thermal-dissipating patterns and thermal-dissipating vias are

illustrated in FIG. 1G, it should be noted that more (or less) levels of the thermal-dissipating patterns and thermal-dissipating vias may be formed in other embodiments.

(38) Still referring to FIG. 1G and also referring to FIGS. 2A-2B, the first thermal-dissipating vias **1961** (and/or the thermal-dissipating vias **1963**) may be arranged in an array within the thermal-dissipating region **1902**. In some embodiments, the first thermal-dissipating vias **1961** (and/or the second thermal-dissipating vias **1963**) are distributed evenly (e.g., having a same pitch) within the thermal-dissipating region **1902**. The first thermal-dissipating vias **1961** (and/or the thermal-dissipating vias **1963**) may each have a circular top-view shape. Although other suitable top-view shape (e.g., oval, rectangular, square, polygonal, etc.) may be possible. The arrangement, size, shape, and number of the thermal-dissipating vias may vary depending on the thermal-dissipating requirements of the product. In some embodiments, the first thermal-dissipating pattern **1962** may include a mesh **1962M** and pads **1962P** formed within openings of the mesh **1962M** from the top view, where the pads **1962P** and the openings of the mesh **1962M** are formed with a one-to-one correspondence. In some embodiments, the boundary of the mesh **1962M** may follow the boundary of the second die **170** (or the boundary of the thermal-dissipating region **1902**) from the top view, and the locations of the pads **1962P** may correspond to the locations of the underlying first thermal-dissipating vias **1961**. The second thermal-dissipating pattern **1964** may include a mesh **1964M** and pads **1964P** that are arranged in a same (or similar) manner as the configuration of the first thermal-dissipating pattern **1962**.

(39) Still referring to FIG. 1G and also referring to FIG. 2C, as seen from the top view, the first thermal-dissipating pattern **1962** overlies the first thermal-dissipating vias **1961**, and the second thermal-dissipating pattern **1964** overlies the second thermal-dissipating vias **1963**. Taking the first thermal-dissipating pattern **1962** and the first thermal-dissipating vias **1961** for example, the width of each column (and/or each row) of the mesh **1962M** may be greater than the diameter of the corresponding group of the first thermal-dissipating vias **1961**. For example, a first group of columns and rows of the first thermal-dissipating vias **1961** are arranged within the columns and rows the mesh **1962M**, respectively. A second group of columns and rows of the first thermal-dissipating vias **1961** may be arranged within the pads **1962P**. In some embodiments, the mesh **1962M** and the pads **1962P** fully (or partially) cover the first thermal-dissipating vias **1961** from the top view. The second thermal-dissipating pattern **1964** and the second thermal-dissipating vias **1963** may have the same (or similar) configuration as the first thermal-dissipating pattern **1962** and the first thermal-dissipating vias **1961**.

(40) Referring to FIG. 1H and with reference to FIG. 1G, the structure of FIG. 1G may be flipped upside-down and disposed on a second temporary carrier **52**. The second temporary carrier **52** may be made of a material such as silicon, polymer, polymer composite, metal foil, ceramic, glass, glass epoxy, tape, or other suitable material for structural support. In some embodiments, an adhesive layer (not shown) is formed on the second temporary carrier **52**, and the third redistribution structure **190** may be attached to the adhesive layer. The adhesive layer may be detached from the second temporary carrier **52** by, e.g., shining UV light in a subsequent carrier de-bonding process. For example, the adhesive layer is a LTHC coating layer or the like.

(41) The first temporary carrier **51** may then be removed by a suitable process, such as etching, grinding, mechanical peeling-off, etc., to accessibly reveal the second surface **110b** of the first redistribution structure **110**. In an embodiment where an adhesive layer (e.g., a LTHC film) is formed between the first temporary carrier **51** and the first redistribution structure **110**, the first temporary carrier **51** is de-bonded by exposing to a laser or UV light. The laser or UV light breaks the chemical bonds of the adhesive layer that binds to the first temporary carrier **51**, and the first temporary carrier **51** may then be de-bonded. Residues of the adhesive layer, if any, may be removed by a cleaning process performed after the carrier de-bonding process. In some embodiments, during the cleaning process, the seed layer (e.g., Ti layer) of the first patterned conductive layer **112** on the second surface **110b** may also be removed.

(42) Referring to FIG. 1I, UBM pads **116** may be formed on the second surface **110b** of the first redistribution structure **110** and may be in physical and electrical contact with the outermost surface **1102b** of the first patterned conductive layer **112**. In some embodiments, the UBM pads **116** in different regions may have different dimensions. For example, a portion of the UBM pads **116** for connecting the subsequently-formed conductive terminals may have a critical dimension greater than a critical dimension of another portion of the UBM pads **116** for connecting the subsequently-mounted electrical device. By way of example and not limitation, a critical dimension of a portion of the UBM pads **116** for connecting the subsequently-formed conductive terminals is about 200  $\mu\text{m}$ , while a critical dimension of a portion of the UBM pads **116** for connecting the subsequently-mounted electrical device is about 55  $\mu\text{m}$ . By way of example and not limitation, a pitch between two adjacent two UBM pads **116** for connecting the subsequently-formed conductive terminals is about 350  $\mu\text{m}$ , while a pitch between two adjacent two UBM pads **116** for connecting the subsequently-mounted electrical device is about 96  $\mu\text{m}$ . Although other values may be possible.

(43) In some embodiments, conductive terminals **210** are formed on some of the UBM pads **116**. The conductive terminals **210** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. The conductive terminals **210** may be solder balls, metal pillars, a ball grid array (BGA), controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold (ENEPIG) technique formed bumps, combination thereof (e.g., a metal pillar having a solder ball attached thereof), or the like. In some embodiments, the conductive terminals **210** include an eutectic material and may include a solder bump, a solder ball, or the like. A reflow process may be performed, giving the conductive terminals **210** a shape of a partial sphere. Alternatively, the conductive terminals **210** may include non-spherical conductive connectors or other shapes.

(44) With continued reference to FIG. 1I, an electrical device **220**, such as an integrated passive device (IPD), is optionally disposed on and electrically coupled to the first redistribution structure **110** through a portion of the UBM pads **116**. Conductive joints **222**, such as solder joints, may be formed between the electrical device **220** and the underlying UBM pads **116**. The conductive joints **222** may include a same material (e.g., solder) as the conductive terminals **210**. The electrical device **220** may be electrically coupled to the first die **130** through the first patterned conductive layer **112** of the first redistribution structure **110**. In some embodiments, a third underfill layer **UF3** may be formed in a gap between the electrical device **220** and the second surface **110b** of the first redistribution structure **110** to surround the conductive joints **222** for protection.

(45) Referring to FIG. 1J and with reference to FIG. 1I, the structure of FIG. 1I may be flipped upside-down to be disposed on a frame **53**, and then the second temporary carrier **52** may be removed to accessibly reveal the third redistribution structure **190**. The removal of the second temporary carrier **52** may be similar to the removal of the first temporary carrier **51** described previously, and thus the detailed descriptions are not repeated. In some embodiments, a patterned dielectric layer **232** is formed on the topmost sublayer of the third dielectric layer **194** and extends across the circuit region **1901** and the thermal-dissipating region **1902**. The patterned dielectric layer **232** may have a different material than the underlying third dielectric layer **194**. The patterned dielectric layer **232** may be or include organic dielectric material, such as an Ajinomoto Buildup Film (ABF), a solder resist film, or the like. In some embodiments, the patterned dielectric layer **232** is referred to as a patterned resist layer.

(46) In some embodiments, a layer of the dielectric material is formed on the third dielectric layer **194**, and then one or more patterning process (e.g., laser drilling, lithography and etching, combinations thereof, or the like) may be performed to form the patterned dielectric layer **232**. In some embodiments, during the patterning process, the underlying third dielectric layer **194** may be partially removed to accessibly reveal the third patterned conductive layer **192** and the second thermal-dissipating pattern **1964**. For example, after the patterning process, first openings **2321** are formed to accessibly reveal the topmost sublayer of the third patterned conductive layer **192**.

corresponding to the circuit region **1901**, and second openings **2322** are formed to accessibly reveal the second thermal-dissipating pattern **1964** corresponding to the thermal-dissipating region **1902**. In some embodiments, the pitch of two adjacent first openings **2321** may be greater than the pitch of two adjacent second openings **2322**. The opening width of the first openings **2321** may be greater than the opening width of the second openings **2322**. By way of example and not limitation, the opening width of the first openings **2321** is about 240  $\mu\text{m}$ , while the opening width of the second openings **2322** is about 200  $\mu\text{m}$ . In alternative embodiments, the opening width of the first openings **2321** may be substantially equal to the opening width of the second openings **2322**. Although other values of the opening widths are fully intended to be included within the scope of the disclosure.

(47) With continued reference to FIG. **1J**, first pre-solder layers **2331** may be formed on the topmost sublayer of the third patterned conductive layer **192** in the first openings **2321**, and second pre-solder layers **2332** may be formed on the second thermal-dissipating pattern **1964** in the second openings **2322**. The first and second pre-solder layers **2331** and **2332** may include materials, e.g., electroless nickel with immersion gold, electroless nickel-phosphorous with immersion gold, electroless nickel with electroless palladium and immersion gold, or the like, and may be formed by printing, dispensing, or the like. In some embodiments, thermal-dissipating bumps **234** may be attached onto the second pre-solder layers **2332** within the second openings **2322**. The thermal-dissipating bumps **234** and the underlying thermal-dissipating vias and patterns are of the same material. The thermal-dissipating bumps **234** may be or may include copper balls, copper studs, copper pillars and other thermal conductive material(s).

(48) The thermal-dissipating features **196** may further include the thermal-dissipating bumps **234** and the underlying second pre-solder layers **2332**. In some embodiments, the structure formed on the frame **53** shown in FIG. **1J** is collectively viewed as a lower package component **101A**. In alternative embodiments, the second pre-solder layers **2332** within the second openings **2322** are omitted, and the thermal-dissipating bumps **234** are directly formed on the second thermal-dissipating pattern **1964** within the second openings **2322**. In alternative embodiments, the thermal-dissipating bumps **234** are omitted. In such case, the lower package component have the second pre-solder layers **2332** formed in the second openings **2322**, and heat generated from the second die **170** may dissipate toward an ambient environment through the second pre-solder layers **2332**.

(49) Referring to FIG. **1K** and with reference to FIG. **1J**, an upper package component **102** may be disposed on and electrically coupled to the lower package component **101A** to form a semiconductor package **10A**. The upper package component **102** may be or may include a memory package or other types of package component which is not limited thereto. In some embodiments, external terminals of the upper package component **102** are disposed on the first pre-solder layers **2331** within the first openings **2321**, and then a reflow process is performed to form conductive joints **1022** electrically and mechanically coupling the upper package component **102** to the lower package component **101A**. In some embodiments, the external terminals of the upper package component **102** and the first pre-solder layers **2331** include a solder material, and the conductive joints **1022** connecting therebetween are solder joints. In some embodiments, no visible interface is formed between the first pre-solder layers **2331** and the conductive joints **1022**.

(50) With continued reference to FIG. **1K** and also referring to FIG. **2D**, the thermal-dissipating bumps **234** in the thermal-dissipating region **1902** may be fully shielded by the upper package component **102**, the conductive joints **1022** formed within the circuit region **1901** may surround the thermal-dissipating bumps **234**. The lateral spacing **LS1** between two adjacent conductive joints **1022** may be greater than the lateral spacing **LS2** between two adjacent thermal-dissipating bumps **234**. The diameter of the conductive joints **1022** may be greater than that of the thermal-dissipating bumps **234**. By way of example and not limitation, the diameter of the conductive joints **1022** is about 350  $\mu\text{m}$ , while the diameter of the thermal-dissipating bumps **234** is about 250  $\mu\text{m}$ . Although other values are fully intended to be included within the scope of the disclosure.

(51) In alternative embodiments, the thermal-dissipating bumps **234** are replaced with one or more thermal-dissipating block (or strip) that may be at least disposed right over hot spot(s) of the second die **170**. It should be noted that the configuration and the number of the conductive joints **1022** and the thermal-dissipating bumps **234** illustrated in FIG. 2D are merely examples and construe no limitation in the disclosure. In some embodiments, the thermal-dissipating bumps **234** are not in physical contact with the upper package component **102**. For example, a vertical spacing **VS1** is formed between the top of each thermal-dissipating bumps **234** and the bottommost surface **102b** of the upper package component **102**, where the vertical spacing **VS1** is non-zero.

(52) After coupling the upper package component **102** to the lower package component **101A**, an underfill layer (not shown) is optionally formed between the upper package component **102** and the lower package component **101A** to surround the conductive joints **1022** and separate the thermal-dissipating bumps **234** from the upper package component **102**. In some embodiments, the lower package component **101A** is formed at wafer level, and a singulation process may be performed to form individual lower package components **101A**, and then the frame **53** may be removed from the lower package components **101A**. After the singulation process, the lower package component **110A** may have a coterminous sidewall including sidewalls of the patterned dielectric layer **232**, the third dielectric layer **194**, the second insulating encapsulation **180**, the second dielectric layer **154**, the first insulating encapsulation **140**, and the first dielectric layer **114**.

(53) The semiconductor package **10A** includes the first die **130** and the second die **170** stacked over the first die **130**, where the active side of the second die **170** faces the back side of the first die **130**, and thus the configuration of the semiconductor package **10A** may be referred to as face-to-back configuration. In some embodiments, the conductive vias of the patterned conductive layers in each redistribution structure (e.g., **190**, **150**, and **110**) may be tapered toward a same direction (e.g., from the conductive joints **1022** toward the conductive terminals **210**). The second die **170** of the semiconductor package **10A** may have a higher thermal output during operation than the first die **130**. The thermal-dissipating features **196** are disposed over and thermally coupled to the second die **170**, and the thermal-dissipating features **196** may propagate heat generated from the second die **170** toward an ambient environment through the first and second thermal-dissipating vias **1961** and **1963**, the first and second thermal-dissipating patterns **1962** and **1964**, the second pre-solder layers **2332**, and the thermal-dissipating bumps **234**.

(54) In some embodiments, the pattern density (e.g., a global pattern density) in the thermal-dissipating area **R1** right above the second die **170** ranges from about 80% to 90% to ensure that the surface area of the thermal-dissipating features **196** is sufficiently large to cool the second die and prevent overheating. The global pattern density, as used herein, may be the total area of the thermal-dissipating features on a photomask divided by the total area of the photomask. In alternative embodiments, the thermal-dissipating bumps **234** are omitted, and the pattern density (e.g., the global pattern density) in the thermal-dissipating area **R1** ranges from about 70% to 80% (or at least 65%).

(55) FIG. 3 is a schematic cross-sectional view illustrating a semiconductor package in accordance with some embodiments. Unless specified otherwise, the materials and the formation methods of the components in these embodiments are essentially the same as the like components, which are denoted by like reference numerals in the embodiments.

(56) Referring to FIG. 3 and with reference to FIG. 1K, the semiconductor package **10B** is similar to the semiconductor package **10A**, except that the lower package component **101B** of the semiconductor package **10B** is free of the thermal-dissipating bumps **234**, the second pre-solder layers **2332**, and the second openings **2322** as included in the semiconductor package **10A**. For example, the patterned dielectric layer **232'** overlying the third dielectric layer **194** of the third redistribution structure **190** fully covers the thermal-dissipating region **1902**. The thermal-dissipating features **196'** including the first and second thermal-dissipating vias **1961** and **1963** and the first and second thermal-dissipating patterns **1962** and **1964** in the thermal-dissipating region

**1902** may be underneath the patterned dielectric layer **232'** and fully embedded in the third dielectric layer **194** of the third redistribution structure **190**. In some embodiments, the pattern density (e.g., the global pattern density) in the thermal-dissipating region **1902** right above the second die **170** ranges from about 70% to 80% (or at least 65%).

(57) FIG. **4** is a schematic cross-sectional view illustrating a semiconductor package FIG. **5A** is a schematic top view illustrating a configuration of a thermal-dissipating pattern, and FIG. **5B** is a schematic top view illustrating a combined configuration of multi-leveled thermal-dissipating patterns, in accordance with some embodiments. Unless specified otherwise, the materials and the formation methods of the components in these embodiments are essentially the same as the like components, which are denoted by like reference numerals in the embodiments.

(58) Referring to FIG. **4** and with reference to FIG. **3**, the semiconductor package **10C** is similar to the semiconductor package **10B**, except for the thermal-dissipating features **196''** in the lower package component **101C** of the semiconductor package **10C**. In some embodiments, the thermal-dissipating features **196''** in thermal-dissipating region **1902** includes multilevel thermal-dissipating patterns (**1961'**, **1962**, **1963'**, and **1964**) stacked upon one another. In some embodiments, the pattern density (e.g., the global pattern density) in the thermal-dissipating region **1902** right above the second die **170** ranges from about 70% to 80% (or at least 65%). In other embodiments, the thermal-dissipating bumps **234** and/or the second pre-solder layers **2332** may be disposed on the topmost thermal-dissipating pattern **1964** for better heat dissipation, and thus the thermal-dissipating bumps **234** and the second pre-solder layers **2332** are illustrated in dash lines to indicate they may (or may not) exist.

(59) With continued reference to FIGS. **4-5A** and also referring to FIGS. **1G** and **2B**, the even-layered thermal-dissipating patterns (**1962** and **1964**) may be similar to the thermal-dissipating patterns (**1962** and **1964**) described in FIGS. **1G** and **2B**. The odd-layered thermal-dissipating patterns (**1961'** and **1963'**) may each have a mesh and pads formed in openings of the mesh. For example, the bottommost thermal-dissipating pattern **1961'** includes a mesh **1961M** and pads **1961P** disposed within openings of the mesh with a one-to-one correspondence from the top view of FIG. **5A**, and bottom surfaces of the mesh **1961M** and the pads **1961P** may be in physical contact with the back surface **170b** of the second die **170** from the cross-sectional view of FIG. **4**. Similarly, the thermal-dissipating pattern **1963'** includes a mesh **1963M** and pads **1963P** disposed within openings of the mesh with a one-to-one correspondence, as shown in FIG. **5A**.

(60) Referring to FIG. **5B** and with reference to FIGS. **2C**, as seen from the top view, the fourth thermal-dissipating pattern **1964** overlies the third thermal-dissipating pattern **1963'**, and the second thermal-dissipating pattern **1962** overlies the first thermal-dissipating pattern **1961'**. The configuration of the thermal-dissipating patterns **1962** and **1961'** may be similar to the thermal-dissipating patterns **1964** and **1963'**. Taking thermal-dissipating patterns **1962** and **1961'** for example, the overlying thermal-dissipating pattern **1962** may fully (or partially) cover the underlying thermal-dissipating pattern **1961'** from the top view. For example, the mesh **1962M** has longer length and width than the length and width of the underlying mesh **1961M**, and the pads **1962P** has greater diameter than the diameter of the underlying pads **1961**.

(61) FIG. **6A** is a schematic cross-sectional view illustrating a semiconductor package **10D**, and FIG. **6B** is a schematic top view illustrating a configuration of the thermal-dissipating feature, the first die, and the second die of the semiconductor package **10D**, in accordance with some embodiments. Unless specified otherwise, the materials and the formation methods of the components in these embodiments are essentially the same as the like components, which are denoted by like reference numerals in the embodiments.

(62) Referring to FIG. **6A** and with reference to FIG. **3**, the semiconductor package **10D** is similar to the semiconductor package **10B**, except that the thermal-dissipating feature **196'** is replaced with the thermal-dissipating feature **296**. The thermal-dissipating feature **296** is a thermally conductive path through which heat that is generated from the second die **170** is dissipated to the ambient. In

some embodiments, the term “thermally conductive” refers to the property of a material to conduct and transfer heat (e.g., allow the heat to flow) from one area to another. For example, the thermal-dissipating feature **296** in the lower package component **101D** is laterally covered by the third dielectric layer **194** of the third redistribution structure **190**, the patterned dielectric layer **232'** overlies the top surface of the thermal-dissipating feature **296**, and the bottom surface of the thermal-dissipating feature **296** is in physical and thermal contact with the back surface **170b** of the second die **170**. The thermal-dissipating feature **296** may be a single thermally conductive layer or a multi-layered thermally conductive structure. In the illustrated embodiments, the pattern density (e.g., the global pattern density) in the thermal-dissipating region **1902** right above the second die **170** is about 100%.

(63) The material of the thermal-dissipating feature **296** may be a metal, such as copper, copper alloy, aluminum, gold, or other suitable metals or metal alloys. In some embodiments, the thermal-dissipating feature **296** includes thermally conductive filler materials that include a filler material from at least one or more of metallic material, semiconductor material, solder material, carbon nano tubes, carbon nano fibers, graphite, etc. The sidewall of the thermal-dissipating feature **296** may be (or may not be) aligned with the sidewall of the second die **170**. In some embodiments, the lateral dimension of the thermal-dissipating feature **296** is less than that of the second die **170**. In alternative embodiments, the thermal-dissipating feature **296** includes a plurality of strips (or blocks) discontinuously distributed on the back surface **170b** of the second die **170**. In other embodiments, the thermal-dissipating bumps **234** and/or the second pre-solder layers **2332** may be disposed on the block of the thermal-dissipating feature **296** for better heat dissipation, and thus the thermal-dissipating bumps **234** and the second pre-solder layers **2332** are illustrated in dash lines to indicate they may (or may not) exist.

(64) With continued reference to FIG. **6A** and also referring to FIG. **6B**, in the top view, an orthographic projection area of the thermal-dissipating feature **296** may be less than an orthographic projection area of the second die **170**, and the orthographic projection area of the second die **170** may be less than an orthographic projection area of the first die **130**. In some embodiments, the boundary line of the thermal-dissipating feature **296** is fully encircled by the boundary line of the second die **170**, from the top view. The boundary line of the second die **170** may be fully (or partially) encircled by the boundary line of the first die **130**. In alternative embodiments, one or more thermal-dissipating feature **296** may be disposed in the location of the thermal-dissipating region **1902** corresponding to a hot spot of the second die **170** for efficient heat dissipation. The top-view shape of the thermal-dissipating feature **296** may be rectangular or square. Alternatively, the thermal-dissipating feature **296** may have other top-view shape (e.g., circular shape, oval shape, polygonal shape, irregular shape, etc.). It should be noted that the number, the shape, and the configuration of the thermal-dissipating feature **296** illustrated herein are merely examples and may be varied depending on thermal dissipation requirements of various semiconductor packages.

(65) FIGS. **7A-7I** are schematic cross-sectional views illustrating various stages of a manufacturing method of a semiconductor package, in accordance with some embodiments. Unless specified otherwise, the materials and the formation methods of the components in these embodiments are essentially the same as the like components, which are denoted by like reference numerals in the embodiments.

(66) Referring to FIG. **7A**, a first redistribution structure **390** including one or more first patterned conductive layer **392** embedded in one or more first dielectric layer **394** may be formed over the temporary carrier **51**. The first patterned conductive layer **392** may be located within the circuit region **3901**. The first patterned conductive layer **392**, the first dielectric layer **394**, and the temporary carrier **51** may be similar to the first patterned conductive layer **112**, the first dielectric layer **114**, and the temporary carrier **51** described in FIG. **1A**, and thus the detailed descriptions are not repeated for the sake of brevity. The first redistribution structure **390** includes a first surface

**390a** and a second surface **390b** opposite to the first surface **390a** and facing the temporary carrier **51**. In some embodiments, the second surface **390b** is the bottommost surface of the first dielectric layer **394**, and a plurality of openings may be formed in the first surface **390a** to accessibly reveal the topmost sublayer of the first patterned conductive layer **392** for further electrical connection. (67) In some embodiments, a thermal-dissipating feature **396** is disposed on the temporary carrier **51** within the thermal-dissipating region **3902** and laterally covered by the first dielectric layer **394** of the first redistribution structure **110**. The thermal-dissipating feature **396** may include a first surface **396a**, a second surface **396b** opposite to the first surface **396a** and facing the temporary carrier **51**, and a sidewall **396c** connected to the first and second surfaces **396a** and **396b** and covered by the first dielectric layer **394**. By way of example and not limitation, the thickness **396H** of the thermal-dissipating feature **396** may be about 40  $\mu\text{m}$ . In some embodiments, the first surface **396a** of the thermal-dissipating feature **396** is substantially leveled (or coplanar) with the first surface **390a** of the first redistribution structure **390**, within process variations.

(68) The thermal-dissipating feature **396** may be placed on the temporary carrier **51** prior to the formation of the first redistribution structure **390**. In some embodiments, the first redistribution structure **390** is formed over the temporary carrier **51**, and then a portion of the first dielectric layer **114** is removed to form a hole, the thermal-dissipating feature **396** is then disposed in the hole. The thermal-dissipating feature **396** may be a dummy die (e.g., a dummy silicon). Alternatively, the material of the thermal-dissipating feature **396** may include a metal or metal alloys. The thermal-dissipating feature **396** may be a single thermally conductive layer or a multi-layered thermally conductive structure. The thermal-dissipating feature **396** may include thermally conductive filler materials for better heat dissipation.

(69) Referring to FIG. 7B, the conductive pillars **160** may be formed on the first surface **390a** of the first redistribution structure **390** with via portions formed in the openings in the first surface **390a** and connecting the pillar portions and the underlying topmost sublayer of the first patterned conductive layer **392**. A die **370'** may be attached to the first surface **396a** of the thermal-dissipating feature **396** through, for example, a connecting film DF1 or any suitable adhesive. In some embodiments, the connecting DF1 may be or may include a die attach film (DAF) or a thermal interface material(s). Alternatively, the connecting film DF1 may be omitted. The die **370'** may be cut from a semiconductor wafer (not shown) and disposed on the thermal-dissipating feature **396** through a pick-and-place process. The die **370'** may be similar to the second die **170**, and the like components are denoted by like reference numerals.

(70) In some embodiments, the die **370'** includes a semiconductor substrate **371** having a front surface **371a** and back surface **371b** on which the connecting film DF1 is disposed, a device layer **373** having semiconductor devices formed in/on the front surface **371a** of the semiconductor substrate **371**, an interconnect structure **374** formed on and electrically coupled to the device layer **373**, die connectors **375** formed on the interconnect structure **374** and electrically coupled to the semiconductor devices in the device layer **373** through the interconnect structure **374**, and a protection layer **276** formed on the interconnect structure **374** and covering the die connectors **375**. At this stage, the die connectors **375** may be buried in the protection layer **276**.

(71) Referring to FIG. 7C and with reference to FIG. 7B, the insulating encapsulation **180** may be formed on the first redistribution structure **390** to cover the conductive pillars **160**, the connecting film DF1, and the die **370**. The material and the formation of the insulating encapsulation **180** may be similar to the second insulating encapsulation **180** described in FIG. 1F, and thus the detailed descriptions are not repeated for the sake of brevity. In some embodiments, a planarization process is performed to level the insulating encapsulation **180**, the conductive pillars **160**, and the die **370'**. During the planarization process, a portion of the protection layer **376** may be removed until the die connectors **375** are accessibly revealed. In some embodiments, the top surface **180t** of the insulating encapsulation **180** is substantially leveled (or coplanar) with the top surfaces **160t** of the conductive pillars **160** and the top surface **370t** of the die **370**, within process variations, where the



top surface **370t** of the die **370** includes the top surface **376t** of the protection layer **376** and the top surfaces **375t** of the die connectors **375**. In some embodiments, the conductive pillars **160** that penetrate through the insulating encapsulation **180** may be referred to as the TIVs **160**.

(72) Referring to FIG. 7D, a second redistribution structure **350** including one or more second patterned conductive layer **352** embedded in one or more second dielectric layer **354** formed on the top surfaces of the insulating encapsulation **180**, the TIVs **160**, and the die **370**. The materials of the second patterned conductive layer **352** and the second dielectric layer **354** are similar to the second patterned conductive layer **152** and the second dielectric layer **154**. For example, the bottommost sublayer of the second patterned conductive layer **352** is in physical and electrical contact with the TIVs **160** and the die connectors **375** of the die **370**. In some embodiments, the topmost sublayer of the second patterned conductive layer **352** includes UBM pads formed on the topmost sublayer of the second dielectric layer **354** for further electrical connection.

(73) Referring to FIG. 7E, conductive pillars **120** may be formed on a portion of the UBM pads of the second patterned conductive layer **352**, the die **130** may be coupled to another portion of the UBM pads of the second patterned conductive layer **352**, and the first underfill layer UF1 may be formed in the gap between the die **130** and the second redistribution structure **350**. The insulating encapsulation **140** may then be formed on the second redistribution structure **350** to cover the conductive pillars **120**, the die **130**, and the first underfill layer UF1. The conductive pillars **120** that penetrate through the insulating encapsulation **140** may be referred to as the TIVs **120**. The forming processes of the insulating encapsulation **140**, the die **130**, and the conductive pillars **120** may be similar to the processes described in FIGS. 1B-1C, and thus the detailed descriptions are not repeated for the sake of brevity. In some embodiments, the top surface **140t** of the insulating encapsulation **140** is substantially leveled (or coplanar) with the top surfaces **120t** of the conductive pillars **120** and the top surface **130t** of the die **130**, within process variations, wherein the top surface **130t** of the die **130** includes the top surfaces of the isolation layer **137** and the TSVs **132**.

(74) Referring to FIG. 7F, a third redistribution structure **310** including one or more third patterned conductive layer **312** embedded in one or more third dielectric layer **314** formed on the top surfaces of the insulating encapsulation **140**, the TIVs **120**, and the die **130**. The materials of the third patterned conductive layer **312** and the third dielectric layer **314** are similar to the patterned conductive layer **112** and the dielectric layer **114**. For example, the bottommost sublayer of the third patterned conductive layer **312** is in physical and electrical contact with the TIVs **120** and the TSVs **132** of the die **130**. In some embodiments, pad portions of the topmost sublayer of the third patterned conductive layer **312** are accessibly revealed by the openings of the topmost sublayer of the third dielectric layer **314** for further electrical connection.

(75) Referring to FIG. 7G, the third redistribution structure **310** may further include UBM pads **316** formed on the topmost sublayer of the third patterned conductive layer **312** for further electrical connection. For example, the conductive terminals **210** are formed over a portion of the UBM pads **316**. In some embodiments, the electrical device **220** is mounted on another portion of the UBM pads **316**. The underfill layer UF3 is optionally formed in a gap between the electrical device **220** and the third redistribution structure **310** to surround the connections therebetween. The formation of the conductive terminals **210** and the step of mounting the electrical device **220** may be similar to the processes described in FIG. 1I, and thus the detailed descriptions are not repeated for the sake of brevity.

(76) Referring to FIG. 7H and with reference to FIG. 7G, the structure of FIG. 7G may be flipped upside-down and placed on the frame **53**. For example, the conductive terminals **210** and/or the electrical device **220** may be attached onto the frame **53**, and the first redistribution structure **390** faces upwardly for further processing. Next, the temporary carrier **51** may be de-bonded to accessibly reveal the first redistribution structure **390**. The de-bonding process of the temporary carrier **51** is similar to the process described in FIG. 1H. Subsequently, a patterned dielectric layer **232'** with openings **2321'** may be formed on the topmost sublayer of the first dielectric layer **394**

and extends across the circuit region **3901** and the thermal-dissipating region **3902**. The openings **2321'** of the patterned dielectric layer **232'** and the underlying openings of the first dielectric layer **394** to accessibly reveal at least a portion of the first patterned conductive layer **392** for further electrical connection. For example, the patterned dielectric layer **232'** is in physical contact with the second surface **396b** of the thermal-dissipating feature **396**. In some other embodiments, the patterned dielectric layer **232'** includes a window (not shown) accessibly exposing the second surface **396b** of the thermal-dissipating feature **396** for better heat dissipation.

(77) Referring to FIG. 7I, the first pre-solder layers **2331** may be formed on the exposed first patterned conductive layer **392** in the openings **2321'** of the patterned dielectric layer **232'**. The upper package component **102** may be disposed on and electrically coupled to the lower package component **101E** to form a semiconductor package **10E**. The external terminals of the upper package component **102** may be disposed on the first pre-solder layers **2331** and reflowed to form the conductive joints **1022** electrically and mechanically coupling the upper package component **102** to the underlying lower package component **101E**. The formation of the first pre-solder layers **2331** and the step of mounting the upper package component **102** may be similar to the processes described in FIGS. 1J-1K. In some embodiments, an underfill layer (not shown) may be formed in a gap between the upper package component **102** and the patterned dielectric layer **232'** to surround the conductive joints **1022**. In some embodiments, the lower package component **101E** is formed at wafer level, and a singulation process may be performed to form individual lower package components **101E** after mounting the upper package component **102**.

(78) The semiconductor package **10E** includes the dies **370** and **130** arranged in a face-to-face configuration since the active sides of the dies **370** and **130** are in proximity to each other. In some embodiments, the conductive vias of the patterned conductive layers in each redistribution structure (e.g., **390**, **350**, and **310**) may be tapered toward a same direction (e.g., from the conductive terminals **210** toward the conductive joints **1022**). The upper die **370** in the lower package component **101E** may have a higher thermal output during operation than the lower die **130**. The thermal-dissipating feature **396** thermally coupled to the upper die **370** may propagate heat generated from the upper die **370** toward an ambient environment. It should be noted that the thermal-dissipating feature **396** in the thermal-dissipating region **3902** may be replaced with any thermal-dissipating feature described somewhere else in the disclosure, depending on product requirements. The thermal-dissipating feature described herein may efficiently route the heat generated within the lower package component to designated areas outside the lower package component.

(79) According to some embodiments, a semiconductor package includes a first die disposed on and electrically coupled to a first redistribution structure, a second die disposed over the first die and laterally covered by a second insulating encapsulation, a second redistribution structure interposed between and electrically coupled to the first and second dies, a third redistribution structure disposed on the second die and opposite to the second redistribution structure, and at least one thermal-dissipating feature. The first die is laterally covered by a first insulating encapsulation, and through substrate vias of the first die are physically connected to the second redistribution structure or the first redistribution structure. The third redistribution structure includes a patterned conductive layer embedded in a dielectric layer, the thermal-dissipating feature is embedded in the dielectric layer of the third redistribution structure and electrically isolated from the patterned conductive layer through the dielectric layer of the third redistribution structure, and the thermal-dissipating feature is thermally coupled to a back surface of the second die.

(80) According to some alternative embodiments, a semiconductor package includes a lower package component and an upper package component stacked over the lower package component. The lower package component includes a lower die and an upper die stacked over and electrically coupled to the lower die, an upper redistribution structure overlying and electrically coupled to the upper die, and at least one thermal-dissipating feature. The upper die has a higher thermal output

during operation than the lower die, the upper redistribution structure includes a patterned conductive layer embedded in a dielectric layer, the thermal-dissipating feature is embedded in the dielectric layer and isolated from the patterned conductive layer of the upper redistribution structure, and the thermal-dissipating feature is in thermal contact with a back surface of the upper die. The upper package component is coupled to the upper redistribution structure through conductive joints.

(81) According to some alternative embodiments, a manufacturing method of a semiconductor package includes: forming a first insulating encapsulation on a first redistribution structure to cover a first die that is coupled to the first redistribution structure; forming a second redistribution structure on the first insulating encapsulation and the first die; coupling a second die to the second redistribution structure; forming a second insulating encapsulation on the second redistribution structure to cover the second die; forming a third redistribution structure on the second insulating encapsulation and the second die; and forming at least one thermal-dissipating feature during forming the third redistribution structure, where the at least one thermal-dissipating feature is embedded in a dielectric layer of the third redistribution structure and electrically isolated from a patterned conductive layer of the third redistribution structure, and the at least one thermal-dissipating feature is thermally coupled to a back surface of the second die.

(82) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

## Claims

1. A semiconductor package, comprising: a first die disposed on and electrically coupled to a first redistribution structure, and the first die laterally covered by a first insulating encapsulation; a second die disposed over the first die and laterally covered by a second insulating encapsulation; a second redistribution structure interposed between and electrically coupled to the first and second dies, and through substrate vias of the first die being physically connected to the second redistribution structure or the first redistribution structure; a third redistribution structure disposed on the second die and opposite to the second redistribution structure, the third redistribution structure comprising a patterned conductive layer embedded in a dielectric layer, the patterned conductive layer comprising a conductive via disposed in a first level of the dielectric layer and a conductive pad disposed over the conductive via and disposed in a second level of the dielectric layer; and at least one thermal-dissipating feature embedded in the dielectric layer of the third redistribution structure and electrically isolated from the patterned conductive layer through the dielectric layer of the third redistribution structure, and the at least one thermal-dissipating feature thermally coupled to a back surface of the second die, wherein the at least one thermal-dissipating feature comprises a thermal-dissipating via disposed in the first level of the dielectric layer and a thermal-dissipating pattern disposed over the thermal-dissipating via and disposed in the second level of the dielectric layer.
2. The semiconductor package of claim 1, wherein the thermal-dissipating via of the at least one thermal-dissipating feature comprises a bottom end connected to the back surface of the second die, a top end wider than the bottom end, and a tilt sidewall connected to the top end and the bottom end.
3. The semiconductor package of claim 1, wherein the at least one thermal-dissipating feature

further comprises: a pre-solder layer overlying the thermal-dissipating pattern, the pre-solder layer comprising a bottom portion laterally covered by the second level of the dielectric layer of the third redistribution structure and a top portion laterally covered by a patterned resist layer which overlies the second level of the dielectric layer of the third redistribution structure.

4. The semiconductor package of claim 3, wherein the at least one thermal-dissipating feature further comprises: thermal-dissipating bumps disposed on the pre-solder layer and protruded from the patterned resist layer, wherein the thermal-dissipating bumps are electrically floating.

5. The semiconductor package of claim 4, wherein the pre-solder layer and the thermal-dissipating bumps are of different materials.

6. The semiconductor package of claim 4, further comprising: a memory package component stacked over and electrically coupled to the third redistribution structure through conductive joints, wherein the conductive joints surround the thermal-dissipating bumps, and the thermal-dissipating bumps interposed between the memory package component and the third redistribution structure are spatially separated from the memory package component.

7. The semiconductor package of claim 1, wherein the at least one thermal-dissipating feature further comprises an additional thermal-dissipating via disposed right over the thermal-dissipating via.

8. The semiconductor package of claim 1, wherein the thermal-dissipating pattern of the at least one thermal-dissipating feature comprises: a first thermal-dissipating pattern comprising a first mesh and first pads disposed within first openings of the first mesh; and a second thermal-dissipating pattern comprising a second mesh and second pads disposed within second openings of the second mesh, the second mesh overlying the first mesh and having a greater dimension than the first mesh, and the second pads overlying the first pads and having a greater dimension than the first pads.

9. The semiconductor package of claim 1, wherein the first redistribution structure comprises a conductive via, and a sidewall of the conductive via of the first redistribution structure and a sidewall of the conductive via of the third redistribution structure are tilt toward a first direction.

10. The semiconductor package of claim 1, wherein in a top view, the at least one thermal-dissipating feature is located within a boundary of the second die, and the boundary of the second die is located within a boundary of the first die.

11. The semiconductor package of claim 1, wherein the second redistribution structure comprises a conductive via, and a sidewall of the conductive via of the second redistribution structure and a sidewall of the conductive via of the third redistribution structure are tilt toward a first direction.

12. A semiconductor package, comprising: a lower package component comprising: a lower die and an upper die stacked over and electrically coupled to the lower die, and the upper die having a higher thermal output during operation than the lower die; an upper redistribution structure overlying and electrically coupled to the upper die, the upper redistribution structure comprising a patterned conductive layer embedded in a dielectric layer; and at least one thermal-dissipating feature embedded in the dielectric layer and isolated from the patterned conductive layer of the upper redistribution structure, the at least one thermal-dissipating feature being in thermal contact with a back surface of the upper die, the at least one thermal-dissipating feature comprising a pre-solder layer embedded in the dielectric layer and thermal-dissipating bumps disposed on the pre-solder layer and protruded from the dielectric layer; and an upper package component stacked over the lower package component and coupled to the upper redistribution structure through conductive joints, wherein the thermal-dissipating bumps are vertically spaced apart from a bottommost surface of the upper package component.

13. The semiconductor package of claim 12, wherein the at least one thermal-dissipating feature comprises: a thermal-dissipating pattern physically connecting the pre-solder layer to the back surface of the upper die and embedded in the dielectric layer of the upper redistribution structure.

14. The semiconductor package of claim 13, wherein a thermal-dissipating via of the thermal-

dissipating pattern is in physical contact with a semiconductor substrate of the upper die.

15. The semiconductor package of claim 12, wherein a lateral spacing between adjacent two of the conductive joints is greater than a lateral spacing between adjacent two of the thermal-dissipating bumps.

16. The semiconductor package of claim 12, wherein a diameter of the respective conductive joint is greater than a diameter of the respective thermal-dissipating bump.

17. A manufacturing method of a semiconductor package, comprising: forming a first insulating encapsulation on a first redistribution structure to cover a first die that is coupled to the first redistribution structure; forming a second redistribution structure on the first insulating encapsulation and the first die; coupling a second die to the second redistribution structure; forming a second insulating encapsulation on the second redistribution structure to cover the second die; forming a third redistribution structure on the second insulating encapsulation and the second die, wherein forming the third redistribution structure comprises forming a patterned conductive layer in a dielectric layer, the patterned conductive layer comprises a conductive via disposed in a first level of the dielectric layer and a conductive pad disposed over the conductive via and disposed in a second level of the dielectric layer; and forming at least one thermal-dissipating feature during forming the third redistribution structure, wherein the at least one thermal-dissipating feature is embedded in the dielectric layer of the third redistribution structure and electrically isolated from a patterned conductive layer of the third redistribution structure, the at least one thermal-dissipating feature is thermally coupled to a back surface of the second die, and the at least one thermal-dissipating feature comprises a thermal-dissipating via disposed in the first level of the dielectric layer and a thermal-dissipating pattern disposed over the thermal-dissipating via and disposed in the second level of the dielectric layer.

18. The manufacturing method of claim 17, wherein forming the at least one thermal-dissipating feature comprises: forming a pre-solder layer on the thermal-dissipating pattern.

19. The manufacturing method of claim 18, wherein forming the at least one thermal-dissipating feature further comprises: forming thermal-dissipating bumps on the pre-solder layer, wherein the thermal-dissipating bumps are protruded from the third redistribution structure, and the pre-solder layer and the thermal-dissipating bumps are of different materials.

20. The manufacturing method of claim 17, wherein forming the thermal-dissipating pattern of the at least one thermal-dissipating feature comprises: forming a mesh and a pad within an opening of the mesh, wherein the pad is directly over the thermal-dissipating via.

---